Chi squared test for independence

1

a H_0: Gender and Test scores are independent. $df = 2$

b H_0: Gender and Job occupation are independent. $df = 3$

c H_0: Sport and Age are independent. $df = 10$

2

a H_0: The preferred sport is independent of age.
H_1: The preferred sport is not independent of age.

b $df = 3$

c $\chi^2 = 12.15$

d $p = 0.00689$

e $p < \alpha$, therefore we reject the null hypothesis H_0. It appears that preferred sport is not independent of age.

3

a H_0: Gender and favourite movie genre are independent.
H_1: Favourite movie genre is not independent of gender.

b $df = 2$

c $\chi^2 = 31.37$

d $\chi^2 > 11.34$, therefore we reject the null hypothesis H_0. It appears that favourite movie genre is not independent of gender.

4

a H_0: The level of window cleanliness is independent of the cleaning product used.
H_1: The level of window cleanliness is not independent of the cleaning product used.

b $df = 6$

c $\chi^2 = 2.32$

d $p = 0.89$

e $p > \alpha$, therefore we do not reject the null hypothesis H_0. It appears that window cleanliness is independent of the cleaning product used.

5

a H_0: Camera technology preference is independent of the client's age.
H_1: Camera technology preference is not independent of the client's age.

b $df = 9$

c 40 people

d $\chi^2 = 57.76$

e $57.76 > 16.92$, therefore we reject the null hypothesis H_0. It appears that camera technology is dependent on the age of the client.

6

a H_0: Blood pressure level is independent of the amount of daily exercise.
H_1: Blood pressure level is not independent of the amount of daily exercise.

b

	< 30	$30 - 60$	> 60	Total
Low bp	12.526	20.632	15.842	49
Average bp	11.248	18.526	14.226	44
High bp	10.226	16.842	12.932	40
Total	34	56	43	133

c $df = 4$

d $\chi^2 = 58.67$

e $p = 0.00$

f $p < \alpha$, therefore we reject the null hypothesis H_0. It appears that blood pressure level is not independent on the amount of daily exercise.

7

- H_0: Voting preference is independent of gender.
  H_1: Voting preference is not independent of gender.
  The significance level $\alpha = 0.05$.
- $df = 2, \chi^2 = 16.20, p = 0.00$
- $p < \alpha$ therefore we reject the null hypothesis. It appears that voting preference is not independent of gender.

8

a H_0: Buying a phone with internet access is independent of age.
H_1: Buying a phone with internet access is not independent of age.

b

	$0- < 12$	$12 - 18$	Over 18	Total
Internet access	5	46	86	137
No internet access	48	7	5	60
Total	53	53	91	197

c $df = 2$

d $\chi^2 = 124.63$

e $124.63 > 9.21$, therefore we reject the null hypothesis H_0. It appears that buying a phone with internet access is not independent on the age of the person.